Introduction to AI Agents

Recap: Large Language Models

Powerful AI models trained on massive amounts of data to learn the complex patterns and rules of human language, allowing them to perform a wide variety of tasks

Large	Language	Model
Trained on **large amounts of data** and have **billions of trainable parameters**	**Deals with text data** (takes input in text and generates output in text)	**Predicts** the next **token**

The Rise of AI Agents

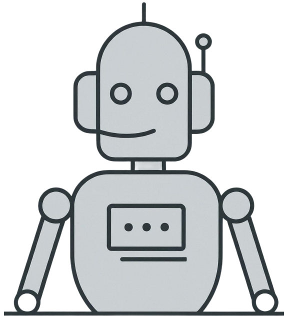
AI that doesn't just respond - it takes action, makes plans, and pursues goals on your behalf.

LEARN

Uses the results of actions to refine and improve its future decisions

PERCEIVE

Receives input from the environment: text, data, web pages, or tool results



AI Agent

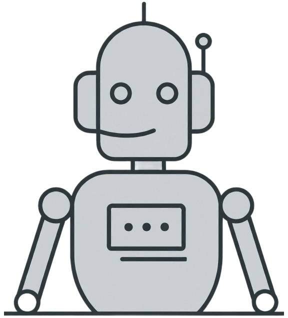
PLAN

Breaks down the goal into smaller steps and decides what to do next

ACT

Executes actions: browsing the web, writing files, calling APIs, running code

Generative AI Agents: Definition



Generative AI agents are an evolving subset of AI Agents that use generative models to perceive their environment, produce novel and relevant outputs, make decisions, and interact with their surroundings to achieve goals with minimal human intervention.

Autonomy	Do not rely on pre-programmed rules or step-by-step instructions and can independently handle complex scenarios relying on reasoning afforded by LLMs
Perception	Use text-based or multimodal inputs and interactions with other systems or users to perceive their environment
Action	Determine the best course of action and execute tasks to fulfil their goals. Task execution is achieved by selecting, sequencing and calling a set of tools available to the agent
Goal-Oriented Behavior	Designed to achieve specific, pre-defined goals by autonomously breaking down complex tasks into smaller steps

AI Agent Design Patterns

Great Learning
POWER AHEAD

Single-Agent Systems

Self-reflective Agents: Iteratively refine their output through self-reflection. Execution concludes when no further improvement is possible.

Planning Agents: Select appropriate tools and define their execution order to accomplish a task, then generate the final output based on the combined tool results.

Multi-Agent Systems

Sequential Pattern: The task is broken down, with dedicated agents handling each subtask consecutively, passing control and results along the chain.

Reflector Pattern: Involves a generator agent creating output and an evaluator agent providing iterative feedback until the output quality is acceptable.

Supervisor Pattern: A supervisor agent plans and delegates subtasks to subordinate agents, then integrates their individual outputs to produce the final result.

Hands-on Illustration:

Multi Agent System with LangGraph


Q
A

Power Ahead!